

Genetic Algorithm

Evolution theory:

Population ↴

P_1	P_2	P_3	P_4
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they face - natural disaster,

diseases and more. they

survive fighting those issues.

Let's say P_4 is the weakest.

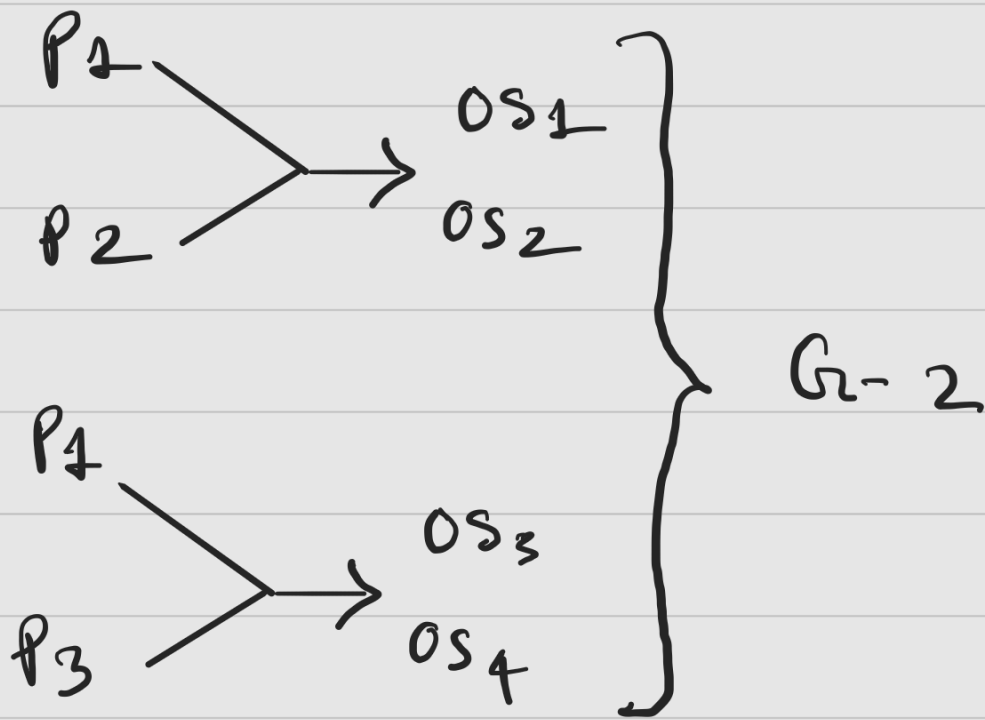
So, P_4 passed away due to

a disease.

P_1	P_2	P_3	P_4
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 } $G-1$

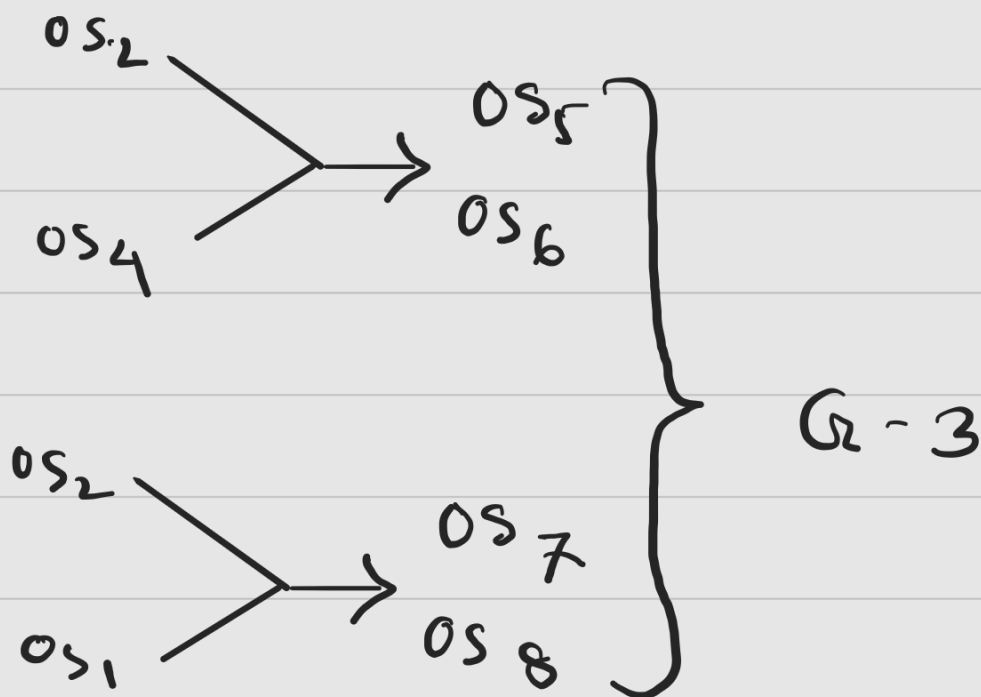
Now, three of them will produce offspring.



As, P_1 , P_2 , and P_3 had good and strong survival traits, the expectation is that those will be passed down to the offsprings as well. However,

among the offsprings, some will be strong and some will be weak. Say, os_3 is the weakest and didn't survive.

now, the offsprings will produce their offsprings



So, this continues and good traits transfer from one generation to another, and we get such a generation which can handle any obstacle flawlessly.

That would be the goal generation.

The evolution theory works pretty much the same way.

In very old times, there

were many diseases such as cholera, plague, TB, and a lot of people used to die. But, in modern times that is not the case, thanks to medicine and our evolution. Then, people learned how to survive and tackle natural disasters.

From this idea we see genetic algorithm.

Let's now talk about the algorithm.

Steps:

i) Selection

Any problem will have a population. So, we have multiple instances of the problem.

We calculate the fitness of each instance and least fit instance is discarded.

So, we select the more/

most fit portion of the population.

ii) Cross over

offspring production is called the crossover.

iii) Mutation

We see variations from parent to child due to mutation.

iv) Goal test

Fitness calculation of

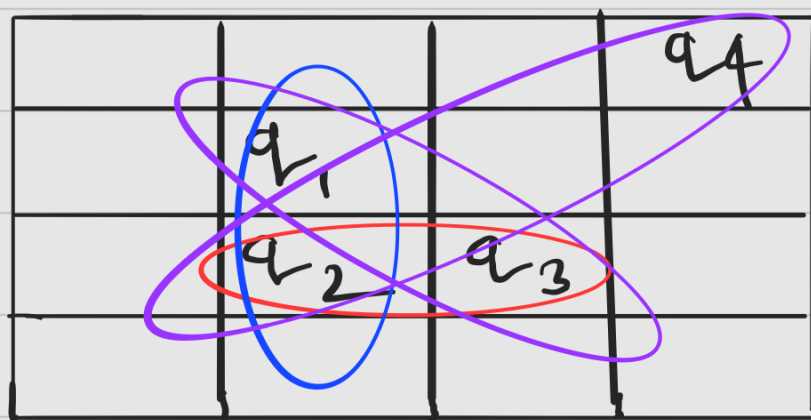
the new population after crossover and mutation to check if it perfectly fits the criteria we want. If it does, then it passes goal test and the algorithm terminates. Otherwise it will repeat.

Let's solve N -queen problem using genetic algorithm.

Game details:

- $N \times N$ board
- Place N queens
- No horizontal/vertical clash
- No diagonal clash

Let's take $N=4$ and see some clashes among the queens.



— horizontal — vertical — Diagonal

One possible solution -

3			a_3	
2	a_1			
1				a_4
0		a_2		

Another solution can be,

3		a_2		
2				a_4
1	a_1			
0			a_3	

How to represent it in a computer? Simply using a string we can do that.

For instance,

0 1 2 3 $\rightarrow$ index (column)
2 0 3 1 $\rightarrow$ row

Here the index represents row number and the value at that index represents the column. So, we have a cell number in 2D, and we place a queen there. Now, let's solve a proper N -queen problem and we set $N = 8$.

Solution:

i) Selection

First, produce a population.

- 1 3 3 0 4 0 1 3 $\rightarrow$ board₁

- 2 1 4 3 2 1 0 2 $\rightarrow$ board₂

- 2 1 6 4 1 3 0 0 $\rightarrow$ board₃

- 1 3 6 3 7 4 4 1 $\rightarrow$ board₄

Now, we need to check the

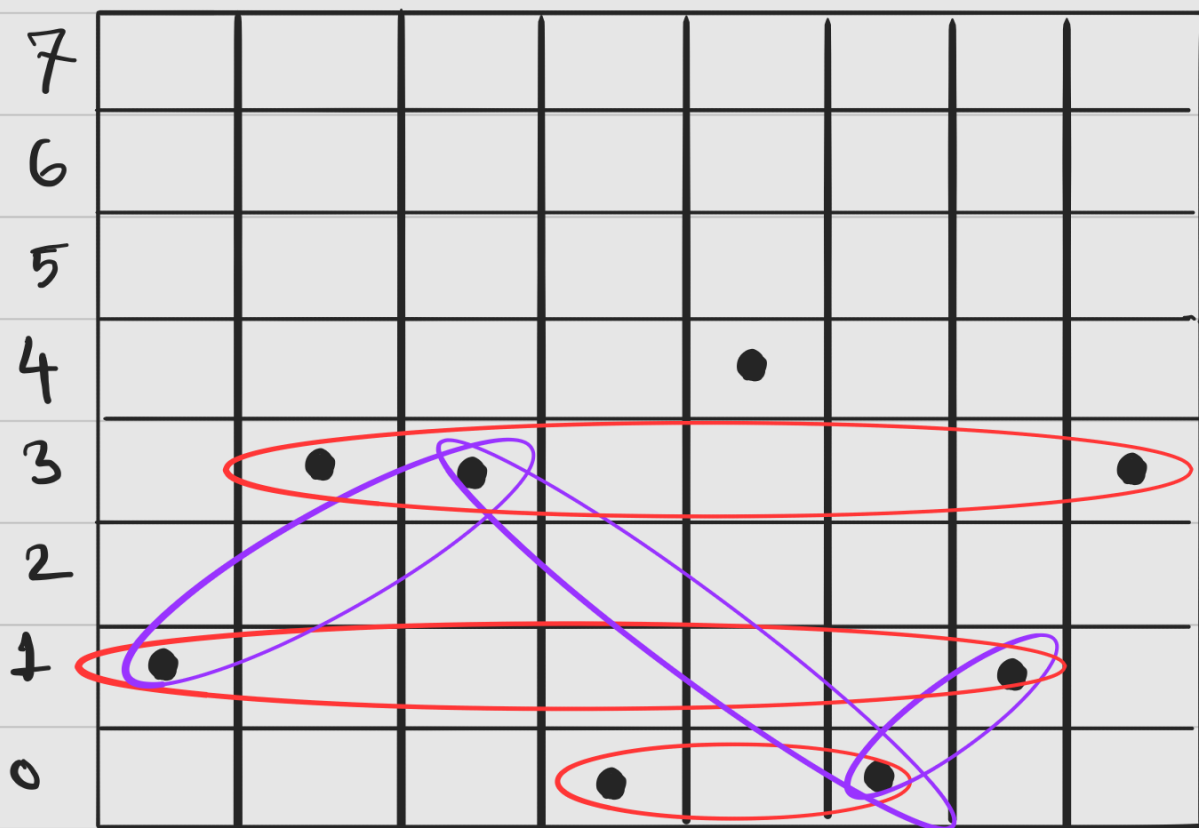
fitness of each board and

discard the worst board.

Let's set the criteria

of fitness - number of
non-clashing pairs.

board-1: 1 3 3 0 4 0 1 3



clash count formula n_{c_2}

$$\text{horizontal} = 3_{c_2} + 2_{c_2} + 2_{c_2}$$

$$\text{diagonal} = 2_{c_2} + 2_{c_2} + 2_{c_2}$$

there will be no vertical
clash as in one row we
place only one queen.

$$\text{so, total clash} = 3 + 1 + 1 + 1 + 1 + 1 \\ = 8$$

$$\text{so, fitness}(\text{board}_1) = \text{total}$$

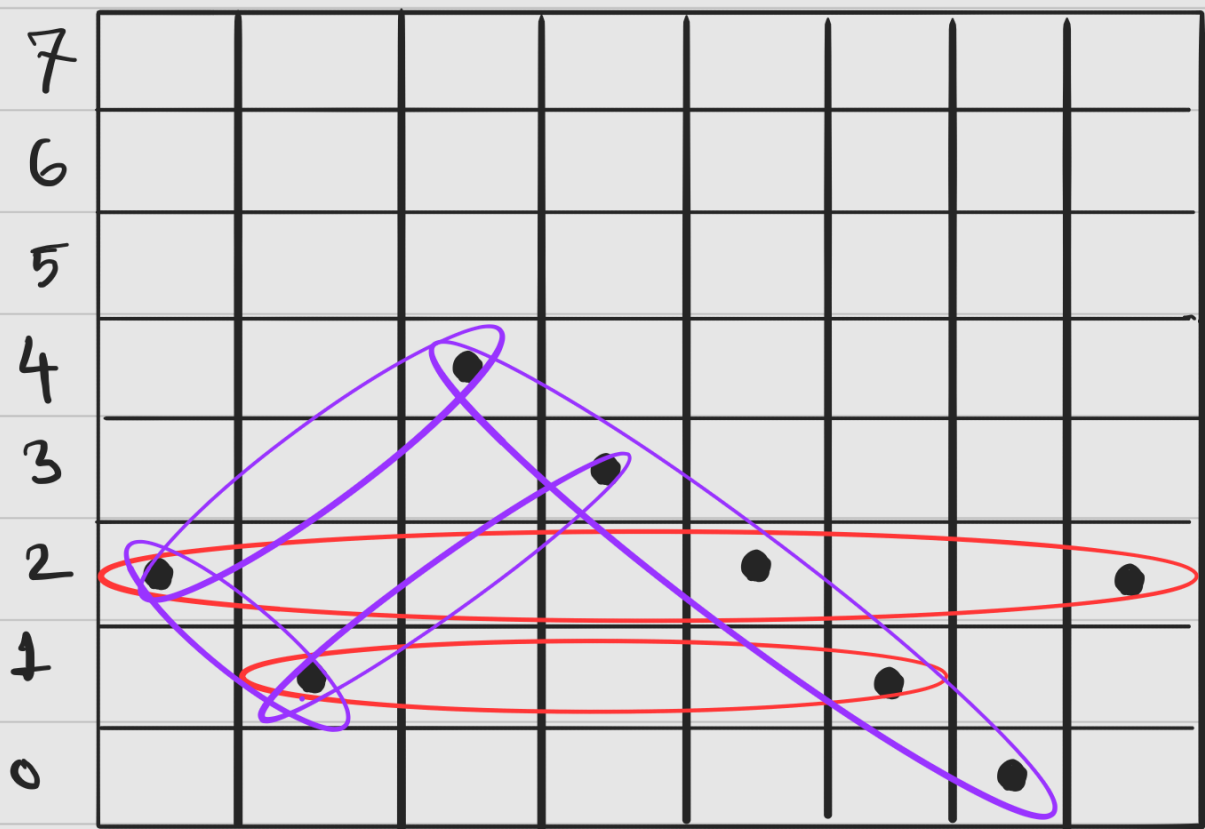
pair- attacking pairs

$$= 8c_2 - 8$$

$$= 28 - 8$$

$$= 20$$

board-2: 2 1 4 3 2 1 0 2



$$\text{horizontal} = 3c_2 + 2c_2$$

$$\text{diagonal} = 2c_2 + 2c_2 + 2c_2 + 5c_2$$

$$\text{so, total clash} = 3 + 1 + 1 + 1 + 1 + 1 + 10 \\ = 17$$

$$\text{so, fitness}(\text{board}_2) = \text{total}$$

pair- attacking pairs

$$= 28 - 17$$

$$= 11$$

So,

$$\begin{array}{cccccccc} 1 & 3 & 3 & 0 & 4 & 0 & 1 & 3 \\ \text{fitness}(\text{board}_1) & = & 20 \end{array}$$

$$\begin{array}{cccccccc} 2 & 1 & 4 & 3 & 2 & 1 & 0 & 2 \\ \text{fitness}(\text{board}_2) & = & 11 \end{array} \quad \times$$

$$\begin{array}{cccccccc} 2 & 1 & 6 & 4 & 1 & 3 & 0 & 0 \\ \text{fitness}(\text{board}_3) & = & 23 \end{array}$$

$$\begin{array}{cccccccc} 1 & 3 & 6 & 3 & 7 & 4 & 4 & 1 \\ \text{fitness}(\text{board}_4) & = & 24 \end{array}$$

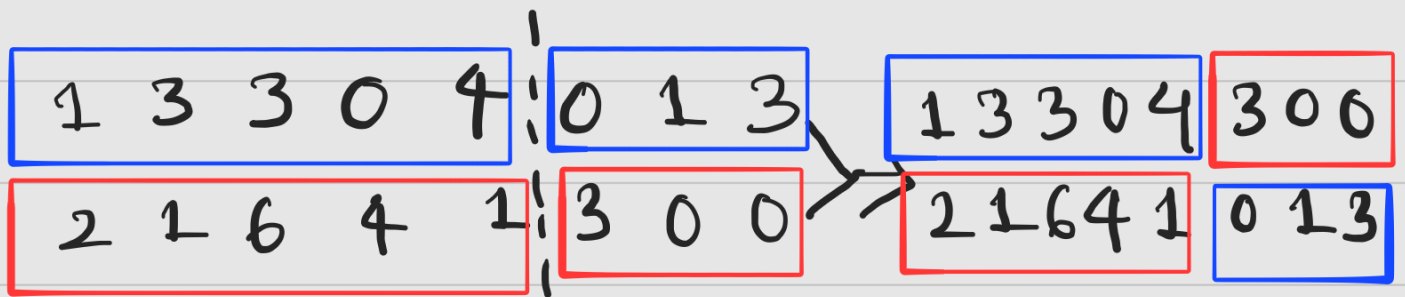
So, worst board is board₂.

We call them chromosomes.

Board₂ has the lowest fitness and we discard it.

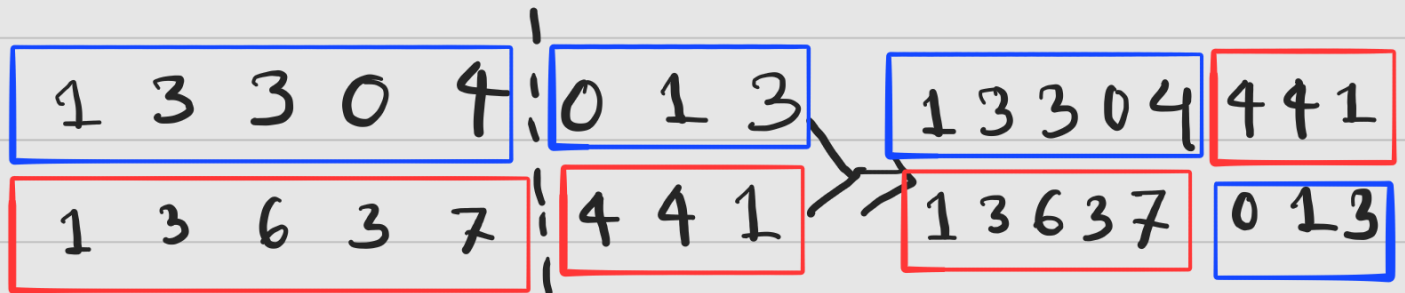
ii) Crossover

Let's choose randomly board₁ & board₃.



For crossover we need to select a point and cut the chromosomes there and combine different parts to produce new offsprings.

Now, randomly choose board₁
& board₄.



iii) Mutation

So, the new offsprings
we have now are -

1 3 3 0 4 3 0 0

2 1 6 4 1 0 1 3

1 3 3 0 4 4 4 1

1 3 6 3 7 0 1 3

Here we will choose a random index and change its value to a random digit.

1 3 3 0 4 3 4 0

2 1 2 4 1 0 1 3

1 3 3 7 4 4 4 1

1 3 6 3 7 4 1 3

Mutation is a very important process of this algorithm.

For instance, a solution of the 4-queen problem

is $= 2031$

3			•	
2	•			
1				•
0		•		

For the solution, all the numbers from 0 to 3 must appear in it.

Say, now the initial population is —

→ $2210, 2210, 1012, 2012$

So, without mutation we can never achieve the final valid solution.

This population has a significant issue which is there is no 3.

iv) Goal test

Here we calculate fitness of the new population and see if the goal is achieved.

Here, $\text{fitness}(\text{board}) = 28$